

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279878

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

KANEYASU; Keisuke

MOVING BODY, ENCRYPTION KEY DELIVERY SYSTEM, ENCRYPTION KEY DELIVERY METHOD, AND PROGRAM

Abstract

A moving body includes an input/output part that receives an encryption key from a transmission apparatus and transmits the encryption key to a reception apparatus; a storage control part that controls storage of the encryption key in a storage part; and a movement control part that moves the moving body to the reception apparatus after receiving the encryption key from the transmission apparatus, wherein the storage control part stores the encryption key in the storage part upon receiving the encryption key from the transmission apparatus and erases the encryption key stored in the storage part in a case where an abnormality is detected.

Inventors:	KANEYASU; Keisuke (Tokyo, JP)
Applicant:	NEC Corporation (Minato- ku, Tokyo, JP)
Family ID:	88918791
Assignee:	NEC Corporation (Minato- ku, Tokyo, JP)
Appl. No.:	18/859647
Filed (or PCT Filed):	May 26, 2022
PCT No.:	PCT/JP2022/021649

Publication Classification

Int. Cl.: **H04L9/08 (20060101)**

U.S. Cl.:

CPC **H04L9/0819 (20130101); H04L9/0852 (20130101);**

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an encryption key delivery technology by a moving body.

BACKGROUND ART

[0002] In the Internet, etc., there is a danger of data being wiretapped during transmission of information (data) from an apparatus on a transmission side to an apparatus on a reception side. To prevent this, there are various encryption technologies that conceal data being transmitted. In many encryption technologies, there is a method where data is encrypted with a key called an encryption key to be transmitted and then decrypted with the same encryption key on the reception side.

Another method is to decrypt data using a decryption key corresponding to the encryption key, held in advance by the reception side. In the following, keys used for encryption and/or decryption of data, including a decryption key, are collectively referred to as encryption keys. Such encryption keys must be shared between the transmission side and the reception side before data transmission.

[0003] Generally, an encryption key is generated by one side and delivered to the other. Therefore, it is important to improve security during encryption key delivery.

[0004] There are various methods for encryption key delivery. For example, quantum cryptography (Quantum cryptography key delivery method) using quantum mechanics. (see, for example, Patent Literature (PTL) 1).

[0005] Quantum cryptography key delivery is a scheme to deliver securely an encryption key by communicating using quantum cryptography. However, a transmission line (transmission medium) is limited to an optical fiber or laser beams, and there are also distance limitations.

[0006] There is a technology that uses a flying body to deliver an encryption key without distance or transmission line restrictions (see, for example, PTL 2). In the technology disclosed in PTL 2, only part of the information necessary to generate the encryption key is temporarily held in the flying body. In other words, the encryption key is shared by a plurality of base stations without having all the information necessary to generate the encryption key in the flying body at the same time.

CITATION LIST

Patent Literature

[0007] [PTL 1] Japanese Unexamined Patent Publication No. 2017-055335 A

[0008] [PTL 2] Japanese Unexamined Patent Publication No. 2022-002379 A

SUMMARY

Technical Problem

[0009] The following analysis has been given by the inventor of the present application.

[0010] In the invention described in PTL 2, instead of the encryption key itself, a plurality of key information necessary to generate the encryption key is sequentially transmitted from the flying body to the plurality of base stations (transmission side and reception side). In the invention described in PTL 2, the encryption key itself is not transmitted from the flying body, so security during transmission and security on the flying body are not considered. The invention described in the PTL 2 is based on a premise that there is a transmission line using optical fiber, etc. between the transmission side and the reception side. The transmission line is used to transmit key information from either the transmission side or the reception side to the other side, and the key distillation process is performed by the side receiving thereof.

[0011] In view of the above-mentioned problems, it is an object of the present disclosure to provide a technique for delivering an encryption key safely, without requiring a transmission line between a transmission side and a reception side, and without distance and transmission line constraints.

Solution to Problem

[0012] According to a first aspect of the present disclosure, there is provided a moving body including: [0013] an input/output part that receives an encryption key from a transmission apparatus and transmits the encryption key to a reception apparatus; [0014] a storage control part that controls storage of the encryption key in a storage part; and [0015] a movement control part that moves the moving body to the reception apparatus after receiving the encryption key from the transmission apparatus, wherein [0016] the storage control part stores the encryption key in the storage part upon receiving the encryption key from the transmission apparatus and erases the encryption key stored in the storage part in a case where an abnormality is detected.

[0017] According to a second aspect of the present disclosure, there is provided an encryption key delivery system including: [0018] the moving body above; [0019] a transmission apparatus; and [0020] a reception apparatus, wherein [0021] the transmission apparatus comprises a transmission apparatus communication part that transmits the encryption key to the moving body, and [0022] the reception apparatus comprises a reception apparatus communication part that receives the encryption key from the moving body.

[0023] According to a third aspect of the present disclosure, there is provided an encryption key delivery method, including: [0024] a reception step for receiving an encryption key from a transmission apparatus; [0025] a storing step for storing the encryption key received in a storage part; [0026] a moving step for moving an own apparatus toward a reception apparatus after storing the encryption key; [0027] an erasure step for erasing the encryption key stored in the storage part in a case where an abnormality of the own apparatus is detected during a movement; and [0028] a transmission step for transmitting the encryption key stored in the storage part to the reception apparatus, upon arriving at a destination area.

[0029] According to a fourth aspect of the present disclosure, there is provided a program causing a computer loaded on a moving body to execute; [0030] a reception step for receiving an encryption key from a transmission apparatus; [0031] a storing step for storing the encryption key received in a storage part; [0032] a moving step for moving an own apparatus toward a reception apparatus after storing the encryption key; [0033] an erasure step for erasing the encryption key stored in the storage part in a case where an abnormality of the own apparatus is detected during a movement; and [0034] a transmission step for transmitting the encryption key stored in the storage part to the reception apparatus, upon arriving at a destination area.

[0035] Further, this program can be recorded on a computer-readable storage medium.

[0036] The storage medium may be a non-transitory one such as semiconductor memory, a hard disk, a magnetic recording medium, an optical recording medium, and the like.

[0037] The present invention can also be realized as a computer program product.

Advantageous Effects of Invention

[0038] According to present disclosure, it is possible to deliver an encryption key safely, without requiring a transmission line between a transmission side and a reception side, and without distance and transmission line constraints.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0039] FIG. 1A is a schematic diagram illustrating one example of an encryption key delivery system of one example embodiment according to the present invention. FIG. 1B is a functional block diagram illustrating one example of a flying body of the key delivery system thereof.

[0040] FIG. 2A and FIG. 2B are a functional block diagram and a hardware configuration diagram of one example of a transmission apparatus according a first example embodiment, respectively.

[0041] FIG. 3A and FIG. 3B are a functional block diagram and a hardware configuration diagram of one example of a reception apparatus according the first example embodiment, respectively.

[0042] FIG. 4A and FIG. 4B are a functional block diagram and a hardware configuration diagram of one example of a flying body according to the first example embodiment, respectively.

[0043] FIG. 5 is a flowchart of one example of an encryption key delivery process in an encryption key delivery system according to the first example embodiment.

[0044] FIG. 6 is a flowchart of one example of an encryption key delivery process in a flying body according to the first example embodiment.

[0045] FIG. 7 is a schematic diagram illustrating one example of a configuration of an encryption key delivery system according to a second example embodiment.

[0046] FIG. 8A and FIG. 8B are a functional block diagram and a hardware configuration diagram of one example of a monitor apparatus according to the second example embodiment, respectively.

[0047] FIG. 9A is a functional block diagram of one example of a flying body according to a third example embodiment. FIG. 9B is a diagram illustrating one example of a log information according to the third example embodiment.

[0048] FIG. 10 is a flowchart of one example of an encryption key delivery process in the flying body according to the third example embodiment.

[0049] FIG. 11 is a schematic diagram illustrating one example of a configuration of an encryption key delivery system according to a fourth example embodiment.

[0050] FIG. 12 is a diagram illustrating an encryption key table according to the fourth example embodiment.

[0051] FIG. 13 is a schematic diagram illustrating one example of a configuration of an encryption key delivery system a according to a variation example of a present disclosure.

[0052] FIG. 14A is a schematic diagram illustrating one example of a configuration of an encryption key delivery system a according to other variation example of a present disclosure. FIG. 14B is a flowchart of one example of an encryption key delivery process in a flying body according thereto.

[0053] FIG. 15 is a schematic diagram illustrating one example of an encryption key table according to other variation example of a present disclosure.

[0054] FIG. 16 is a schematic diagram illustrating one example of a configuration of an encryption key delivery system a according to other variation example of a present disclosure.

EXAMPLE EMBODIMENTS

[0055] Hereinafter, an example embodiment of the present disclosure is described with reference to drawings. Note that reference signs in the drawings provided here are for the sake of convenience for each element as an example to promote better understanding, and are not intended to limit the present disclosure to the illustrated modes. Connection lines between blocks in the drawings referred to in the following description can be both bidirectional and unidirectional. A unidirectional arrow schematically shows the main flow of a signal (data) and does not exclude bidirectionality.

[0056] A program is executed by a computer apparatus, and the computer apparatus is provided with, for example, a processor, a storage device, an input device, a communication interface, and a display device as needed. Further, the computer apparatus is configured to be able to perform wired or wireless communication with an internal device therein or with an external device (including a computer) via the communication interface. Although input/output connection points of each block in the drawings have ports or interfaces, these are not illustrated. Further, in the following description, “A and/or B” means at least one of A and B.

[0057] An example embodiment, which is an example of the present disclosure, is explained. The present example embodiment realizes an encryption key delivery system using a flying body as a moving body, which is difficult to wiretap. In the present example embodiment, an encryption key is loaded on the flying body and transmitted from a source device (transmission side) to a destination device (reception side). At this time, security is ensured using sensor signals from various sensors inside and outside the flying body. In the present example embodiment, an

apparatus on the transmission side of the encryption key is referred to as a transmission apparatus and an apparatus on the reception side of the encryption key as a reception apparatus.

[0058] FIG. 1A illustrates an example of an encryption key delivery system **910** of an example embodiment of a present disclosure. As illustrated in FIG. 1A, the encryption key delivery system **910** is provided with a transmission apparatus **100**, a reception apparatus **200**, and a flying body **300**.

[0059] In the example in FIG. 1A, the transmission apparatus **100** is disposed at a base on the land (on the ground). The reception apparatus **200** is disposed on, for example, a marine vessel on the sea. As the flying body **300**, for example, a drone, which is an example of an unmanned aerial vehicle (UAV) capable of autonomous navigation, is used. Disposed locations of the transmission apparatus **100** and the reception apparatus **200** are not limited thereto.

[0060] In the encryption key delivery system **910** of the present example embodiment, an encryption key **500** is transmitted from the transmission apparatus **100** to the flying body **300**, the flying body **300** flies to the reception apparatus **200**, and the reception apparatus **200** receives the encryption key **500** from the flying body **300**. Upon detecting an abnormality, the flying body **300** erases the encryption key **500** that it holds. Not only in a case of device abnormality, but also in a case where the encryption key **500** is wiretapped due to an unexpected circumstance or where the encryption key **500** is threatened with non-delivery, the flying body **300** determines that an abnormality has occurred and erases the encryption key **500**. The unexpected circumstance may be, for example, contact with another flying body or a failure of the flying body **300** itself.

[0061] To realize this, the flying body **300** is provided with an input/output part **311**, a storage control part **312**, and a movement control part **315**. The input/output part **311** receives the encryption key **500** from the transmission apparatus **100** and also transmits the encryption key **500** to the reception apparatus. The storage control part **312** stores the encryption key **500** in an encryption key storage part **321** upon receiving it from the transmission apparatus **100**. The movement control part **315** moves the flying body **300** from the transmission apparatus **100** to the reception apparatus **200**. The storage control part **312** then erases the encryption key **500** stored in the encryption key storage part **321** if an abnormality is detected during movement.

[0062] Note that the present example embodiment does not assume that a file (information itself) to be transmitted and received is actually loaded on the flying body **300**. This is because, considering a flight speed of the flying body **300**, real-time communication is not possible. Since the encryption key **500** only needs to be shared before communication, a delivery of the encryption key **500** is not required to be real-time compared to general communication.

First Example Embodiment

[0063] Hereinafter, a first example embodiment of the present disclosure is described. According to the first example embodiment, in an encryption key delivery system **910**, a flying body **300** is provided with an abnormality detection means to detect an abnormality that occurs during movement. The following is a detailed description of each apparatus.

[Transmission Apparatus]

[0064] FIG. 2A illustrates a functional block diagram of a configuration related to the present example embodiment of a transmission apparatus **100**. As illustrated in FIG. 2A, the transmission apparatus **100** is provided with a communication part **111**, a control part **112**, an authentication part **113**, an encryption key storage part **121**, and an authentication information storage part **122**.

[0065] The encryption key storage part **121** stores an encryption key **500** therein. The encryption key **500** is generated in the transmission apparatus **100** or received from an external apparatus.

[0066] The information part authentication storage **122** stores authentication information therein. The authentication information is information to prove that the flying body **300** is authorized to deliver the encryption key **500**. The authentication information is set in advance for each flying body **300**, and the transmission apparatus **100**, the reception apparatus **200**, and the flying body **300** hold the same information. The authentication information is stored in the authentication

information storage part **122** in advance.

[0067] The communication part **111** transmits and receives data to and from an external apparatus. In the present example embodiment, the communication part **111** transmits and receives the encryption key **500** and the authentication information to and from flying body **300**.

[0068] As described above, a laser beam, for example, is used to transmit the encryption key **500** from the transmission apparatus **100** to the flying body **300**. The communication using a laser beam can have quantumness. Therefore, the communication part **111** transmits the encryption key **500** to the flying body **300**, for example, with a quantum communication or a quantum cryptography communication.

[0069] The communication part **111** may transmit and receive the authentication information by a laser beam as well as the encryption key **500**. The communication part **111** may transmit and receive the authentication information by short-range radio communication such as Bluetooth (registered trademark) or Wi-Fi (registered trademark).

[0070] The authentication part **113** authenticates the flying body **300**. Upon receiving the authentication information via the communication part **111**, the authentication part **113** collates the received authentication information with the authentication information stored in the authentication information storage part **122** according to an instruction of the control part **112**. After collation, the authentication part **113** notifies the control part **112** of the collation results. The authentication part **113** sets the collation result to “authentication success” in a case where the received authentication information and the stored authentication information match, and to “authentication failure” in a case where they do not match.

[0071] The control part **112** controls an operation of the entire transmission apparatus **100** and also controls transmission and reception of the encryption key **500** via the communication part **111**. Concretely, the control part **112** reads the encryption key **500** from the encryption key storage part **121** upon receiving notification of authentication success from the authentication part **113** and transmits it to the flying body **300** via the communication part **111**.

[0072] FIG. 2B illustrates a hardware configuration diagram of a configuration related to the present example embodiment of the transmission apparatus **100**. As illustrated in the diagram, the transmission apparatus **100** is provided with a CPU **131**, a storage device **132**, and a communication device **133**.

[0073] The CPU **131** loads a program stored in a non-volatile area of the storage device **132** into a work area and executes it to realize each of above functions and to comprehensively control the entire transmission apparatus **100**. One or more processors such as a micro processing unit (MPU) may be used instead of the CPU **131**.

[0074] The storage device **132** is configured with a memory such as a ROM (Read Only Memory) or a RAM (Random Access Memory). The storage device **132** stores information such as programs for executing each function and communication parameters for communication. In the present example embodiment, each of the above storage parts is constructed in the storage device **132**.

[0075] In addition to memories such as a ROM and a RAM, the storage device **132** may be provided with storage media such as an SSD (Solid State Drive), a flexible disk, a hard disk, an optical disk, a CD-ROM, a CD-R, a magnetic tape, a non-volatile memory card, a DVD, etc. The storage device **132** may be provided with a plurality of memories, etc.

[0076] The communication device **133** realizes data transmission and reception. In the present example embodiment, the communication device **133** is provided with, for example, a laser beam output apparatus (luminous source) that outputs a laser beam and an optical wireless communication apparatus having a laser beam reception apparatus, etc. A laser diode (LD), a light emitting diode (LED), etc. that outputs a visible laser beam may be used as the luminous source.

[0077] In addition, the communication device **133** may also use, radio waves, sound waves, etc., as well as visible light to transmit and receive data. Concretely, the communication device **133** may be provided with a transmitter/receiver in accordance with other wireless communication schemes,

such as Bluetooth, Wi-Fi, NFC (Near Field Communication), etc.

[Reception Apparatus]

[0078] FIG. 3A illustrates a functional block diagram of a configuration related to the present example embodiment of the reception apparatus **200**. As illustrated in the diagram, the reception apparatus **200** is provided with a communication part **211**, a control part **212**, an authentication part **213**, an encryption key storage part **221**, and an authentication information storage part **222**, and an authentication information storage part **222**.

[0079] The control part **212** controls an operation of the entire reception apparatus **200** and also controls transmission and reception of the encryption key **500** via the communication part **211**. Concretely, the control part **212** generates an encryption key transmission request upon receiving notification of authentication success from the authentication part **213** and transmits it to the flying body **300** via the communication part **111**. The control part **212** stores an encryption key **500** in the encryption key storage part **221** upon receiving it in response to an encryption key transmission request. At this time, a signal, meaning storage completion, may be transmitted to the flying body **300**.

[0080] Since each of the other functions of the reception apparatus **200** is basically the same as the function of the same name in the transmission apparatus **100**, they are not described here. In a case where it is necessary to distinguish between the functions of the transmission apparatus **100** and the reception apparatus **200**, each function is referred to as a transmission apparatus communication part **111**, a reception apparatus communication part **211**, a transmission apparatus control part **112**, a reception apparatus control part **212**, a transmission apparatus authentication part **113**, and a reception apparatus authentication part **213**, respectively.

[0081] FIG. 3B illustrates a hardware configuration diagram of a configuration related to the present example embodiment of the reception apparatus **200**. As illustrated in the diagram, the reception apparatus **200** is provided with a CPU **231**, a storage device **232**, and a communication device **233**. Since each configuration of the reception apparatus **200** is basically the same as that of the transmission apparatus **100** of the same name, they are not described here.

[Flying Body]

[0082] Next, the flying body **300**, which delivers the encryption key **500** from the transmission apparatus **100** to the reception apparatus **200**, is described. In the present example embodiment, a case of using a drone as the flying body **300** will be used as described above.

[0083] FIG. 4A illustrates a functional block diagram of a configuration related to the present example embodiment of the flying body **300**. The flying body **300** of the present example embodiment is provided with the input/output part **311**, the storage control part **312**, the movement control part **315**, and the encryption key storage part **321**, as illustrated in FIG. 1A, as well as an abnormality detecting part **313**, an authentication request part **314**, an authentication information storage part **322**, and an authentication information storage part **322**.

[0084] The encryption key storage part **321** stores the encryption key **500** therein. The encryption key **500** is received from the transmission apparatus **100**.

[0085] The authentication information storage part **322** stores the authentication information of the flying body **300** therein.

[0086] The input/output part **311** performs input/output of data to/from an external apparatus. The data to be input/output includes the encryption key **500**, authentication information, etc.

[0087] The input/output part **311** receives the encryption key **500** from the transmission apparatus **100** via wireless communication and transmits the encryption key **500** to the reception apparatus **200** via wireless communication as described above. The input/output part **311** transmits authentication information to the transmission apparatus **100** and the reception apparatus **200**. The input/output part **311** may have a function that allows data to be transmitted and received only under predetermined conditions, such as time and location.

[0088] As described above, transmission and reception of the encryption key **500** between the

flying body **300** and the transmission apparatus **100** or the reception apparatus **200** may be done, for example, by quantum communication or quantum cryptography communication using a laser beam.

[0089] The authentication information may be transmitted and received via a laser beam as well as the encryption key **500**. The authentication information may also be transmitted and received via Bluetooth, Wi-Fi, or other short-range radio communication.

[0090] The authentication request part **314** requests authentication to the external apparatus. In the present example embodiment, the authentication request part **314** requests authentication to the transmission apparatus **100** and the reception apparatus **200**. The authentication request part **314** transmits authentication information stored in the authentication information storage part **322** to request authentication. Upon receiving a notification from the movement control part **315**, described below, that an own apparatus (the flying body **300**) has entered an area in which it can communicate with the external apparatus subject to the authentication request (hereinafter referred to as a communicable area), the authentication request part **314** requests authentication.

[0091] The abnormality detecting part **313** monitors a status of the flying body **300** and transmits an abnormality detecting signal to the storage control part **312** in a case where an abnormality is detected.

[0092] The abnormality detecting part **313** collects, for example, sensor detection values transmitted from a sensor **334**, described below. The abnormality detecting part **313** outputs an abnormality detecting signal in a case where the collected sensor detection values have abnormality (abnormal value(s)). Whether or not the value is abnormal is determined by whether or not it exceeds an allowable range predetermined for each sensor. The allowable range is predetermined by, for example, a threshold value.

[0093] The abnormality detecting part **313** may also analyze the collected sensor detection values to determine presence or absence of abnormality. The abnormality detecting part **313** may also detect, for example, that the flying body **300** is no longer able to fly, has crash-landed or crashed, has a dead battery or program trouble, etc.

[0094] The storage control part **312** controls storing the encryption key **500** in the encryption key storage part **321**, as described above. In the present example embodiment, the storage control part **312**, for example, acquires the encryption key **500** via the input/output part **311** and stores it in the encryption key storage part **321**. Upon receiving an encryption key transmission request, the storage control part **312** reads the encryption key **500** from the encryption key storage part **321** and transmits the encryption key **500** to the request source via the input/output part **311**. The storage control part **312** further erases the encryption key **500** stored in the encryption key storage part **321** upon receiving an abnormality detecting signal from the abnormality detecting part **313**.

[0095] The movement control part **315** navigates the flying body **300** to its destination as described above. In the present example embodiment, the movement control part **315** navigates the flying body **300** from the transmission apparatus **100** to the reception apparatus **200**. In the present example embodiment, upon receiving the encryption key **500** from the transmission apparatus **100**, the movement control part **315** controls the navigation device (see below) and moves the flying body **300** to the reception apparatus **200**.

[0096] A hardware configuration of the flying body **300** is described below. FIG. **4B** illustrates a hardware configuration diagram of a configuration related to the present example embodiment of the flying body **300**. As illustrated in the diagram, the flying body **300** is provided with a CPU (Central Processing Unit) **331**, a storage device **332**, a communication device **333**, a sensor **334**, and a navigation device **335**.

[0097] The CPU **331** loads a program stored in a non-volatile area of the storage device **332** into a work area and executes it to realize each of above functions and to comprehensively control the entire flying body **300**. One or more processors such as a micro processing unit (MPU) may be used instead of the CPU **331**.

[0098] The storage device **332** is configured with a memory such as a ROM (Read Only Memory) or a RAM (Random Access Memory). The storage device **332** stores information such as programs for executing each function and communication parameters for communication. In the present example embodiment, each of the above storage parts is constructed in the storage device **332**.

[0099] In addition to memories such as a ROM and a RAM, the storage device **332** may be provided with storage media such as an SSD (Solid State Drive), a flexible disk, a hard disk, an optical disk, a CD-ROM, a CD-R, a magnetic tape, a non-volatile memory card, a DVD, etc. The storage device **332** may be provided with a plurality of memories, etc.

[0100] The communication device **333** communicates with an external apparatus. In the present example embodiment, the communication device **333** performs wireless communication with, for example, the transmission apparatus **100** and the reception apparatus **200**. As described above, the encryption key **500** is transmitted and received via a laser beam. Therefore, the communication device **333** is provided with a laser input apparatus and a laser beam reception apparatus. The communication device **333** may also be provided with a near-distance communication device that transmits and receives authentication information.

[0101] The sensor **334** is provided in each part of the flying body **300** and detects its status. The sensor **334** is set up to detect (observe) all events that affect a flight of the flying body **300**, such as vibration, contact, temperature change, voltage, position (latitude, longitude, and height), movement speed (flight speed), travel time, contact with other devices (other objects), interference, etc. The sensor **334** may be capable of detecting whether or not data is being transmitted to or received from other apparatuses. In the present example embodiment, the sensor **334** includes, for example, a gyro sensor, acceleration sensor, barometric sensor, ultrasonic sensor, geomagnetic sensor (electronic compass), camera, GPS device, battery level sensor, etc.

[0102] These detection values are output to the abnormality detecting part **313** and the movement control part **315**, which are used to detect abnormality and control navigation using the navigation device **335**.

[0103] The navigation device **335** navigates the flying body **300** according to a control of the movement control part **315**. In the present example embodiment, the navigation device **335** is provided with, for example, a propeller and a motor to rotate the propeller. The movement control part **315** controls the navigation device **335** according to a navigation program stored in the storage device **332** and an output from the sensor **334** to make the flying body **300** navigate. The movement control part **315**, for example, adjusts an output of the motor and controls a speed of the propeller based on information obtained from the sensor **334**.

[Encryption Key Delivery Method]

[0104] The following is an example flow of an encryption key delivery process using the encryption key delivery system **910** with the above configuration. FIG. 5 illustrates a flow diagram of an example of the encryption key delivery process by the encryption key delivery system **910** in the present example embodiment.

[0105] Upon reaching a communicable area of the transmission apparatus **100**, the flying body **300** transmits authentication information to the transmission apparatus **100** (step **S1301**) and requests authentication.

[0106] Upon receiving authentication the information as an authentication request from the flying body **300**, the transmission apparatus **100** performs authentication (step **S1101**). Here, the authentication part **113** collates the received authentication information with the authentication information stored in the authentication information storage part **122**. If both match, the authentication part **113** notifies the control part **112** of the authentication success.

[0107] Upon receiving the authentication success notification, the control part **112** reads the encryption key **500** from the encryption key storage part **121** and transmits it to the flying body **300** of an authentication request source via the communication part **111** (step **S1102**).

[0108] If the authentication is unsuccessful, the transmission apparatus **100** does not transmit the

encryption key **500** to the flying body **300** of the authentication request source. In this case, a message indicating authentication failure, etc. may be transmitted to the flying body **300** of the authentication request source.

[0109] Upon receiving the encryption key **500**, the flying body **300** stores it (step **S1302**) and starts moving (step **S1303**).

[0110] Upon reaching a communicable area of the reception apparatus **200** without abnormality being detected (step **S1304**; No), the flying body **300** terminates its movement (step **S1305**). Then, the flying body **300** transmits the authentication information to the reception apparatus **200** (step **S1306**) and requests authentication.

[0111] Upon receiving the authentication information as an authentication request from flying body **300**, the reception apparatus **200** performs authentication (step **S1201**). Here, authentication part **213** collates the received authentication information with the authentication information stored in the authentication information storage part **222**. If both match, the authentication part **213** notifies the control part **212** of the authentication success.

[0112] Upon receiving notification of authentication success, the control part **212** generates an encryption key transmission request and transmits thereof to the flying body **300** of the authentication request source via communication part **111** (step **S1202**).

[0113] In a case where the authentication is unsuccessful, the reception apparatus **200** does not transmit the encryption key transmission request to the flying body **300** of the authentication request source. In this case, a message indicating authentication failure may be transmitted to the flying body **300** of the authentication request source.

[0114] Upon receiving the encryption key transmission request from the reception apparatus **200**, the flying body **300** transmits the stored encryption key **500** to the reception apparatus **200** of the request source (step **S1307**). Then the flying body **300** erases the stored encryption key **500** (step **S1308**).

[0115] On the other hand, if an abnormality is detected during movement (step **S1304**; Yes), the flying body **300** erases the stored encryption key **500** (step **S1308**) and terminates the process.

[0116] After receiving the encryption key **500** in step **S1307**, the reception apparatus **200** communicates to the transmission apparatus **100** that the reception apparatus **200** has received the encryption key **500**, using radio wave or the like and starts to use the encryption key **500**. After receiving encryption key **500**, the cryptographic algorithm used in the reception apparatus **200** is not limited. The encryption key **500** transmitted and received is not used for encryption and decryption as it is, but may be used by being converted using an algorithm stored in advance in the transmission apparatus **100** and the reception apparatus **200**.

[0117] Next, an example of a process in the flying body **300** at a time of encryption key delivery is described. FIG. **6** illustrates a flowchart of an encryption key delivery process by the flying body **300** of the present example embodiment.

[0118] The movement control part **315** determines whether the flying body **300** has reached within a communicable area of the transmission apparatus **100** (step **S1401**). It does not matter which determination method is used. Various existing determination methods can be used. For example, a laser beam for distance measurement may be transmitted to the communication part **111**, and if a distance value obtained is equal to or less than a predetermined value, the movement control part **315** may determine that it has reached therewithin. The movement control part **315** may determine using the captured images obtained by a camera. If a location of the transmission apparatus **100** is known, the movement control part **315** may determine using, for example, GPS functionality.

[0119] In a case where the flying body **300** reaches the communicable area, the authentication request part **314** makes an authentication request. Here, as described above, the authentication request part **314** transmits the authentication information stored in the authentication information storage part **322** to the transmission apparatus **100** (step **S1402**).

[0120] Upon receiving the encryption key **500** within a predetermined period of time from the

transmission apparatus **100** in response to the authentication request (step **S1403**), the storage control part **312** stores the encryption key **500** in the encryption key storage part **321** (step **S1404**). [0121] If the flying body **300** does not receive the encryption key **500** within a predetermined period of time, for example, if it receives a message of the authentication failure or if it receives no reply, it terminates the process.

[0122] Next, after finishing storing the encryption key **500**, the movement control part **315** causes the flying body **300** to start moving (step **S1405**). Here, the movement control part **315** controls the navigation device **335** to move the flying body **300** to a communicable area of the reception apparatus **200** (step **S1407**).

[0123] During movement, upon detecting an abnormality (step **S1406**), the abnormality detecting part **313** outputs an abnormality detecting signal to the storage control part **312**. Upon receiving the abnormality detecting signal, the storage control part **312** erases the encryption key **500** stored in the encryption key storage part **321** (step **S1412**) and terminates the process.

[0124] On the other hand, if the flying body **300** reaches the communicable area without detecting abnormality (step **S1407**), the movement control part **315** terminates the movement thereof (step **S1408**). The method for determining whether or not it reaches the communicable area of the reception apparatus **200** is the same as that used for the transmission apparatus **100**.

[0125] Then, the authentication request part **314** makes an authentication request to the reception apparatus **200**. Here, the authentication request part **314** transmits the authentication information stored in the authentication information storage part **322** to the reception apparatus **200** (step **S1409**).

[0126] Upon receiving an encryption key transmission request within a predetermined period of time from the reception apparatus **200** in response to the authentication request (step **S1410**), the storage control part **312** reads the encryption key **500** from the encryption key storage part **321** and transmits it to the reception apparatus **200** via the input/output part **311** (step **S1411**). Then, the storage control part **312** erases the encryption key **500** in the encryption key storage part **321** (step **S1412**) and terminates the process.

[0127] If the flying object **300** does not receive the encryption key transmission request within the predetermined period of time, for example, if it receives a message of the authentication failure, or if it receives no reply, it moves to step **S1412**.

[0128] As explained above, in the encryption key delivery system **910** of the present example embodiment, a flying body **300** is used to deliver the encryption key **500** between the transmission apparatus **100** and the reception apparatus **200**. The flying body **300** is provided with an abnormality detecting function, and if an abnormality is detected in the flying body **300**, the encryption key **500** to be delivered is erased.

[0129] For the flying body **300**, a flying body capable of unmanned and autonomous navigation, such as a drone, is used. The flying body **300** is not susceptible to physical contact as long as it is flying at a sufficient altitude. Thus, the encryption key delivery system **910** in the present example embodiment uses the flying body **300** to deliver the encryption key **500**. This allows the system **910** to achieve high security key delivery with little risk of wiretapping, for example, even between two points where no transmission line is drawn, without any restrictions on distance or transmission line.

[0130] For example, as illustrated in FIG. **1A**, if a transmission apparatus **100** is disposed at a base on the land and a reception apparatus **200** is disposed on a marine vessel sailing on the sea, it is difficult to connect both apparatuses with physical cables because the marine vessel moves. In a situation where such a transmission line cannot be secured, the encryption key delivery system **910** of the present example embodiment is particularly useful.

[0131] The encryption key delivery system **910** of the present example embodiment also uses wireless communication for transmitting and receiving the encryption key **500** between the transmission apparatus **100** and the flying body **300**, or the reception apparatus **200** and the flying

body **300**. This means that encryption key **500** can be transmitted and received without a connector connection, eliminating the need to add a new hardware to the flying body **300** and reducing costs. [0132] The encryption key delivery system **910** in the present example embodiment uses a laser beam for the wireless communication. The laser beam has a small diffusion. Therefore, at a time transmitting and receiving the encryption key **500** between the transmission apparatus **100** and the flying body **300**, or between the reception apparatus **200** and the flying body **300**, the transmission is focused on a very small area. This allows for a smaller transmitting/receiving interface area for each of the flying body **300**, the transmission apparatus **100**, and the reception apparatus **200**. Therefore, in the encryption key delivery system **910** of the present example embodiment, attacks on a transmitting and receiving interface are extremely difficult, and encryption key delivery with high security can be realized.

[0133] For example, consider a case where the flying body **300** is wiretapped on its way to a destination. It is very difficult to access a small laser beam interface of the flying body **300**, which is during navigation, to obtain information. Therefore, in order to obtain the encryption key **500**, an attacker may capture the flying body **300**.

[0134] However, in the encryption key delivery system **910** of the present example embodiment, in a case where the flying body **300** contacts an obstacle during navigation, the abnormality detecting part **313** detects changes in vibration, radio waves, temperature, etc., and outputs an abnormality detecting signal. Then, the storage control part **312** immediately erases the encryption key **500** stored in the encryption key storage part **321**. Therefore, according to the encryption key delivery system **910** of the present example embodiment, even if the attacker captures the flying body **300**, the encryption key **500**, which was in the flying body **300**, cannot be obtained because it has been erased. The encryption key delivery system **910** of the present example embodiment can realize highly secure encryption key delivery from this point of view as well.

[0135] A use of a laser beam allows a transmission line to be visually confirmed, even in wireless communication, making it easy to set an area range and to set a position for transmission and reception with high accuracy. Therefore, according to the encryption key delivery system **910**, an encryption key delivery, which is difficult to intercept and wiretapping, with a high success probability can be realized. Furthermore, according to the encryption key delivery system **910**, by using a laser beam, a quantum cryptography can be used for transmission and reception of the encryption key **500** between the transmission apparatus **100** and the flying body **300** or the reception apparatus **200** and the flying body **300**. This enables an encryption key delivery having the same security as a case where the encryption key **500** is delivered directly between the transmission apparatus **100** and the reception apparatus **200** using quantum cryptography.

[0136] In the present example embodiment, during communication between the flying body **300** and the transmission apparatus **100** and/or the reception apparatus **200**, the transmission apparatus **100** and/or reception apparatus **200** may output a laser beam therefrom, or the flying body **300** may output a laser beam to these apparatuses. Different transmission and reception mechanisms may be used in a communication between the transmission apparatus **100** and the flying body **300**, and in a communication between the reception apparatus **200** and the flying body **300**, respectively.

[0137] In the present example embodiment, the flying body **300** side determines whether or not it has entered the communicable area. However, this may be done by the transmission apparatus **100** and the reception apparatus **200** sides.

[0138] In this case, the transmission apparatus **100** and the reception apparatus **200** are provided with sensors and the like to detect the flying body **300**, and these detect that the flying body **300** has entered the communicable area. Then, the transmission apparatus **100** or the reception apparatus **200** transmits an authentication information transmission request to the flying body **300**.

[0139] In the present example embodiment, the flying body **300** immediately erases the stored encryption key **500** upon transmitting the encryption key **500** to the reception apparatus **200**. However, it is not limited thereto. For example, it may be configured to erase the stored encryption

key **500** after the flying body **300** confirms that the reception apparatus **200** has received thereof. [0140] In this case, upon receiving the encryption key **500**, the reception apparatus **200** transmits a reception notice to the flying body **300**. The flying body **300** waits for a receipt of the reception notice and then erases the stored encryption key **500**. The flying body **300** may be configured to transmit the encryption key **500** multiple times to the reception apparatus **200** until it receives a reception notice.

[0141] In a case where the flying body **300** has erased the stored encryption key **500**, it may notify the transmission apparatus **100** and/or the reception apparatus **200** thereof. Also, in a case where the authentication is unsuccessful in step **S1201** above, the reception apparatus **200** may transmit a message that means authentication failure in plain text to the transmission apparatus **100** therefrom.

Second Example Embodiment

[0142] Next, a second example embodiment of the present disclosure is described. The encryption key delivery system **910** in the first example embodiment uses the sensor **334** mounted on the flying body **300** to detect an abnormality during delivery. In the present example embodiment, a flying body **300** is further monitored by an external apparatus to detect an abnormality of the flying body **300**.

[0143] FIG. **7** illustrates an example of an encryption key delivery system **920** in the present example embodiment. As illustrated in the figure, the encryption key delivery system **920** of the present example embodiment has basically the same configuration as that of the first example embodiment. That is, the encryption key delivery system **920** is provided with a transmission apparatus **100**, a reception apparatus **200**, and a flying body **300**. The encryption key delivery system **920** of the present example embodiment is further provided with a monitor apparatus **400**. The following description of the present example embodiment focuses on the monitor apparatus **400**, which is a different configuration from the first example embodiment.

[0144] The monitor apparatus **400** monitors the flying body **300** and its surroundings from the outside. The monitor apparatus **400** is configured with, for example, a camera and/or a radio wave reception and transmission apparatus.

[0145] The monitor apparatus **400**, for example, in a case where it is a camera, is installed so that the flying body **300** is in its field of view. In a case where it is a radio wave reception and transmission apparatus, the monitor apparatus **400** is installed so that the flying body **300** enters in a direction of radio wave emission.

[0146] FIG. **8A** and FIG. **8B** illustrate a functional block of the monitor apparatus **400** and a hardware configuration of the monitor apparatus **400**, respectively.

[0147] As illustrated in these drawings, the monitor apparatus **400** is provided with a communication part **411**, an information acquisition part **412**, and an abnormality detection part **413**. The monitor apparatus **400** is provided with a CPU **431**, a storage device **432**, a communication device **433**, and an information collection device **434**.

[0148] The information collection device **434** collects information on and near the flying body **300**. The information collection device **434** is an imaging unit in a case where it is a camera, or a radio wave receiver/transmitter in a case where it is a radio wave reception and transmission apparatus.

[0149] The information acquisition part **412** controls an operation of the information collection device **434** and passes information it collects at predetermined time intervals to the abnormality detection part **413**. In the present example embodiment, the information acquisition part **412** collects image or reflected wave information.

[0150] The abnormality detection part **413** analyzes images or reflected waves acquired by the information acquisition part **412** to determine presence or absence of an abnormality in the flying body **300**. If the abnormality detection part **413** determines that an abnormality exists, i.e., if it detects an abnormality, it transmits an abnormality detecting signal to the flying body **300** via the communication part **411**.

[0151] The communication part **411**, the CPU **431**, the storage device **432**, and the communication

device **433** are similar to configurations of the same name of, for example, the transmission apparatus **100** of the first example embodiment. Therefore, the explanation is omitted.

[0152] The flow of the encryption key delivery process of the present example embodiment is basically the same as that of the first example embodiment. However, in the present example embodiment, in step **S1406**, the storage control part **312** erases the encryption key **500** stored in the encryption key storage part **321** even if the abnormality detecting signal is received via the input/output part **311** as well as from the abnormality detecting part **313**.

[0153] The monitor apparatus **400** may be installed on the ground, or it may be a satellite or other floating object. It may be, for example, a flying body that flies alongside the flying body **300**. The transmission apparatus **100** or the reception apparatus **200** may also be provided with a monitor apparatus **400**.

[0154] As explained above, the present example embodiment is provided with the same configuration as the first example embodiment and thus has the same effect as the first example embodiment. Furthermore, the encryption key delivery system **920** of the present example embodiment is provided with the monitor apparatus **400** that monitors the flying body **300** and the state of its surroundings from the outside. If the monitor apparatus **400** determines that the flying body **300** has abnormality, the flying body **300** also erases the encryption key **500** during delivery. Therefore, according to this embodiment, the abnormality of the flying body **300** can be discovered with a higher degree of accuracy, and the delivery of encryption key **500** with higher security can be realized.

[0155] In the present example embodiment, the flying body **300** does not have to be provided with the abnormality detecting part **313**.

[0156] In the present example embodiment, the monitor apparatus **400** transmits an abnormality detecting signal directly to the flying body **300** in a case where it detects an abnormality. However, it is not limited to this. For example, the monitor apparatus **400** may be configured to transmit the abnormality detecting signal to the transmission apparatus **100** and/or the reception apparatus **200**. In this case, the abnormality detecting signal is transmitted from these apparatuses to the flying body **300**.

Third Example Embodiment

[0157] Next, a third example embodiment of the present disclosure is described.

[0158] In each of the above example embodiments, the flying body **300** is monitored in real time during movement to detect the abnormality of the flying body **300**. In the present example embodiment, sensor detection values during movement is accumulated as log information. Then, for example, after the flying body **300** reaches a vicinity of the reception apparatus **200**, this log information is further analyzed to detect presence or absence of abnormality.

[0159] Configuration of the encryption key delivery system, the transmission apparatus **100**, and the reception apparatus **200**, and a hardware configuration of the flying body **300** are basically the same as those of the first example embodiment, and therefore explanation is omitted here.

[0160] An example of a functional block of the flying body **300** of the present example embodiment is illustrated in FIG. **9A**. As illustrated in the drawing, the flying body **300** of the present example embodiment is provided with a log information storage part **323** in addition to the configuration of the flying body **300** of the first example embodiment. The log information storage part **323** is constructed in the storage device **332**.

[0161] The abnormality detecting part **313** of the present example embodiment stores the collected sensor detection values as log information in the log information storage part **323** in correspondence with a detection time (or collection time) in addition to performing a process of the first example embodiment.

[0162] The abnormality detecting part **313** collects and stores not only the sensor detection value of the sensor **334** that detects a state of the flying body **300**, but also access history to the storage device **332** as log information.

[0163] In addition, the abnormality detecting part **313** of the present example embodiment performs a log information verification process after the flying body **300** reaches a communicable area of the reception apparatus **200**. The log information verification process is a process analyzing log information and verifying whether there is an abnormality. In a case where an improper log is detected in the log information verification process, the abnormality detecting part **313** determines that an abnormality exists and outputs an abnormality detecting signal. Similar to the first example embodiment, the storage control part **312** erases the encryption key **500** in the encryption key storage part **321** upon receiving the abnormality detecting signal.

[0164] For example, if log information **600** illustrated in FIG. **9B** is recorded as an access log to the encryption key storage part **321**, the abnormality detecting part **313** determines that an abnormality exists and outputs the abnormality detecting signal.

[0165] In a case where the flying body **300** receives the encryption key **500** from the transmission apparatus **100**, after it stores (in) the encryption key **500** in the encryption key storage part **321**, no access will be occurred until it passes the encryption key **500** to the reception apparatus **200**.

However, in the log information **600**, the read (out) is recorded twice. Therefore, the abnormality detecting part **313** determines that there is an improper access, i.e., there is an abnormality.

[0166] FIG. **10** illustrates an example of a process at the time of encryption key delivery by the flying body **300** of the present example embodiment. It is the same as in the first example embodiment until the flying body **300** reaches the reception apparatus **200** (step **S1408**).

[0167] In the present example embodiment, the abnormality detecting part **313** then performs a log information verification process (step **S3101**). If the abnormality detecting part **313** determines that there is an abnormality in the log information verification process (step **S3102**; Yes), it outputs an abnormality detecting signal. Then, in response, the storage control part **312** erases the encryption key **500** in the encryption key storage part **321** (step **S1412**).

[0168] On the other hand, if the abnormality detecting part **313** finds an abnormality during the log information verification process (step **S3102**; No), it moves the process to step **S1409** and continues the same process as the first example embodiment. The log information verification process may be performed at any time between the end of the movement and the transmission of the encryption key.

[0169] As explained above, the present example embodiment is provided with the same configuration as the first example embodiment and thus has the same effect as the first example embodiment. In addition, the present example embodiment also verifies presence or absence of abnormality using accumulated information. This allows for the delivery of 500 encryption keys with higher security.

[0170] Even if an attacker attempts wiretapping in a way that disables an abnormality detecting function, for example, by accessing a laser beam interface without contacting an airframe of the flying body **300**, the present example embodiment can confirm the presence or absence of the wiretapping by checking the transmission and reception log of the encryption key **500**.

[0171] In the present example embodiment, the log information is stored in the flying body **300** and a log information verification process is performed therein but is not limited to this. It may be configured that, during the movement of the flying body **300**, the log information of the flying body **300** may be transmitted to the transmission apparatus **100** or the reception apparatus **200**, which accumulate the log information and perform the log information verification process. In this case, in a case where the abnormality is detected, an abnormality detecting signal is transmitted from these apparatuses to the flying body **300**.

[0172] In addition, the present example embodiment, like the second example embodiment, may be provided with a monitor apparatus **400**. In this case, the monitor apparatus **400** may also accumulate the collected information as the log information and perform the log information verification process. It may be configured that, the log information accumulated by the monitor apparatus **400** may be transmitted to the transmission apparatus **100** or the reception apparatus **200**,

which performs the log information verification process. In this case, in a case where the abnormality is detected, also, an abnormality detecting signal is transmitted from these apparatuses to the flying body **300**.

[0173] The reception apparatus **200** may combine a result of monitoring by the monitor apparatus **400** with the log information to make a more detailed determination of the presence or absence of abnormality. For example, a satellite camera or radio wave is used to track the flying body **300** until it passes data to the reception apparatus **200**. As a method of detecting an abnormality at this time, if an unscheduled object approaches the flying body **300** and contacts it, the reception apparatus **200** is notified. The reception apparatus **200** may determine that the flying body **300** is being abused if an abnormality is confirmed against the log information (flight log or contact log) of flying body **300**. If an unscheduled object contacts the flying body **300**, it may be assumed to have been wiretapped, and an abnormality detection signal may be transmitted to the flying body **300** to erase the encryption key **500** in the flying body **300**.

[0174] In addition, the reception apparatus **200** may query the flight log of the flying body **300**, sensor detection value log, transmission and reception log, etc., to the transmission apparatus **100** and/or the monitor apparatus **400** to determine if an unscheduled event has occurred. Then, if the reception apparatus **200** may determine that an unscheduled event has occurred, it may have the encryption key **500** in the flying body **300** erased for fear of wiretapping.

[0175] In addition, the reception apparatus **200** may determine whether a flight time is appropriate. In this case, the reception apparatus **200** may have the encryption key **500** in the flying body **300** erased if the flying body **300** has not arrived within a pre-determined time.

[0176] In this case, for example, upon transmitting the encryption key **500** to the flying body **300**, the transmission apparatus **100**, at the same time, transmits a message to the reception apparatus **200** indicating that it has transmitted the encryption key **500**. If the flying body **300** does not arrive within a predetermined time from receipt of the message, the reception apparatus **200** generates an abnormality detecting signal. At this time, the reference apparatus **200** may search for the flying body **300** and transmit the abnormality detecting signal at the timing when it generates the abnormality detecting signal. At a time when authentication information is transmitted from the flying body **300**, the reception apparatus **200** may reply with an abnormality detecting signal without performing authentication.

[0177] The storage control part **312** of the flying body **300** that receives the abnormality detecting signal erases the encryption key **500** without transmitting it to the reception apparatus **200**.

[0178] The flying body **300** may determine whether a flight time is appropriate. That is, in a case where a predetermined time has elapsed since receipt of the encryption key **500**, the storage control part **312** erases the encryption key **500** in the encryption key storage part **321**.

[0179] The period of time until erasure can be specified at a time of individual delivery. For example, upon receipt of the encryption key **500**, the flying body **300** may be configured to also receive a specification of the period of time until erasure, from transmission apparatus **100**. The storage control part **312** erases the encryption key **500** after this period passes.

Fourth Example Embodiment

[0180] Next, a fourth example embodiment of the present disclosure is described. In each of the above example embodiments, one flying body **300** is navigated from the transmission apparatus **100** to the reception apparatus **200**. On the other hand, in the present example embodiment, a plurality of flying bodies **300** is navigated from the transmission apparatus **100** to the reception apparatus **200**, and a plurality of encryption keys **500** is delivered.

[0181] An example of the encryption key delivery system **940** of the present example embodiment is illustrated in FIG. **11**. As illustrated in FIG. **11**, the encryption key delivery system **940** is provided with a transmission apparatus **100** and a reception apparatus **200**, similar to the first example embodiment. The encryption key delivery system **940** of the present example embodiment is further provided with a plurality of flying bodies **300**.

[0182] The respective configurations of the transmission apparatus **100**, the reception apparatus **200**, and the flying body **300** are basically the same as in the first example embodiment. The following description of the present example embodiment focuses on its different configuration from that of the first example embodiment.

[0183] Each flying body **300** has its own unique authentication information. The transmission apparatus **100** of the present example embodiment stores the encryption key **500** in correspondence with authentication information in the encryption key storage part **121**, as encryption key table **140**. In the present example embodiment, the authentication information storage part **122** is not required.

[0184] In the encryption key table **140**, the encryption key **500** to be delivered is registered, corresponding to the authentication information of each flying body **300**. An example of the encryption key table **140** of the present example embodiment is illustrated in FIG. **12**.

[0185] As illustrated in FIG. **12**, each record in the encryption key table **140** has an authentication information recording part **141** and an encryption key recording part **142**. The authentication information recording part **141** stores the authentication information of each flying body **300** authorized to deliver the encryption key **500**. The encryption key recording part **142** stores the encryption key **500** to be delivered with the flying body **300** identified by the corresponding authentication information.

[0186] The reception apparatus **200** may also have an encryption key table having records with the same entries. The authentication information recording part stores authentication information of all the flying bodies **300** authorized to deliver the encryption key **500**, respectively.

[0187] In the authentication process of step **S1101** in FIG. **5**, the authentication part **113** collates all the authentication information stored in the authentication information recording part **141**. If there is any matching authentication information, it transmits the encryption key **500**, which is registered in correspondence with the authentication information matched, to flying body **300**.

[0188] If none of the authentication information stored in the authentication information recording part **141** matches the authentication information, the authentication part **113** determines “authentication failure”.

[0189] In addition, in the authentication process of step **S1201** in FIG. **5**, the reception apparatus **200** collates all the authentication information stored in the authentication information recording part. If any of the authentication information matches, it transmits an encryption key transmission request to the flying body **300**. Then, the reception apparatus **200** stores the encryption key **500** transmitted from the flying body **300**, in response, in the encryption key recording part in correspondence with the matched authentication information.

[0190] This allows supply of the encryption keys **500** to be increased within the same time period, according to the present example embodiment.

[0191] The same encryption key **500** may be loaded and delivered in a plurality of flying bodies **300**. In this case, the delivery of the encryption key **500** is more reliable.

[0192] In addition, the present example embodiment may include a dummy in the **500** encryption keys to be transmitted. A portion of the encryption key **500** to be transmitted may be a dummy, or the entire encryption key **500** delivered by a particular flying body **300** may be a dummy. The dummy key (bit) is not used as an encryption key. In this case, too, the encryption key **500** to be transmitted and received may be converted and used by the transmission apparatus **100** and the reception apparatus **200** using an algorithm stored in advance. Furthermore, determination of whether or not it is a dummy may be made using an algorithm stored in advance by the transmission apparatus **100** and the reception apparatus **200**. These methods can reduce the risk of wiretapping, capture, etc.

[0193] Note that the present example embodiment may be combined with the second example embodiment.

Variation Example 1

[0194] In each of the above example embodiments and variation examples, during transmission and

reception of the encryption key **500** between the transmission apparatus **100** and the flying body **300** or the reception apparatus **200** and the flying body **300**, the flying body **300** is in a gliding state and is transmitting/receiving the encryption key **500** by wireless communication. For example, the communication therebetween may be wired communication, as illustrated in FIG. **13**. FIG. **13** illustrates an example of an encryption key delivery system **910** similar to the first example embodiment.

[0195] In this case, the flying body **300** lands in a vicinity of the transmission apparatus **100** or the reception apparatus **200** at the time of transmitting and receiving the encryption key **500**. Then, as illustrated in this drawing, the two are connected by a physical cable to transmit and receive the encryption key **500**.

[0196] In this case, the communication device **133** of the transmission apparatus **100**, the communication device **233** of the reception apparatus **200**, and the communication device **333** of the flying body **300** are provided with wired communication connectors. Furthermore, instead of the communication devices **133**, **233**, and **333**, each apparatus may be provided with an interface that inputs/outputs data via a medium and transmits/receives the encryption key **500** via the medium.

[0197] As mentioned above, in a case where the transmission apparatus **100**, the reception apparatus **200** and the flying body **300** are provided with multiple data communication and input/output interfaces, the exchange of the encryption key **500** between the transmission apparatus **100** and the flying body **300**, and the exchange of the encryption key **500** between the reception apparatus **200** and the flying body **300** may be in different communication schemes, respectively.

[0198] In this case, the transmission and reception of information about authentication between the flying body **300** and the transmission apparatus **100** or the flying body **300** and the reception apparatus **200** may also be by wired cable. In each example embodiment and variation example, wired cables may be used only for transmitting and receiving information related to authentication.

Variation Example 2

[0199] In each of the above example embodiments and variation examples, a case of delivery of an encryption key **500** from one transmission apparatus **100** to one reception apparatus **200** is explained as an example. However, it is not limited thereto. For example, as illustrated in FIG. **14A**, the same encryption key **500** may be delivered by the same flying body **300** to a plurality of reception apparatuses **200** in sequence.

[0200] In this variation example, the movement control part **315** of the flying body **300** causes the flying body **300** to fly toward each of the reception apparatuses **200** in a predetermined order. Flight routes may be pre-programmed.

[0201] In this variation example, at the time of receiving the encryption key **500**, the storage control part **312** receives from the transmission apparatus **100** the number of the reception apparatus **200** of delivery destination.

[0202] FIG. **14B** illustrates a process flow at the time of encryption key delivery by the flying body **300** in this case. Note that the number of the reception apparatus **200** of delivery destination is N (N is an integer greater than or equal to 2).

Also, n is a counter.

[0203] The flying body **300** sets the counter n to an initial value of 1 (step **S4501**). Then, flying body **300** executes the process from step **S1401** to step **S1411** in FIG. **6**, and executes the encryption key **500** delivery process to the reception apparatus **200** scheduled to deliver the nth (step **S4502**).

[0204] Thereafter, it is determined whether or not the encryption key **500** has been erased (step **S4503**). For example, if an abnormality is detected during the delivery process in step **S4502**, the encryption key **500** has already been erased. In this case, subsequent delivery is not possible, and the process is terminated as is.

[0205] On the other hand, if the encryption key **500** has not been erased, the flying body **300**

repeats the delivery process, incrementing the counter by 1 until the last (Nth) reception apparatus **200** (steps **S4504**, **S4505**).

[0206] After the delivery process is completed up to the last reception apparatus **200**, the storage control part **312** erases the encryption key **500** stored in the encryption key storage part **321** (step **S4506**) and terminates the process.

[0207] There may be a plurality of transmission apparatuses **100**. In this case, the flying body **300** goes around a plurality of transmission apparatuses **100** in a predefined order and receives a different encryption key **500** from each transmission apparatus **100**. Then, the flying body **300** delivers those encryption keys to the one reception apparatus **200**.

[0208] Furthermore, there may be a plurality of transmission apparatuses **100** and a plurality of the reception apparatuses **200**, respectively. Then, a plurality of encryption keys **500** may be received from one transmission apparatus **100**, or a plurality of encryption keys **500** may be transmitted to one reception apparatus **200**.

[0209] In this case, the flying body **300** receives the encryption key **500** together with, for example, information identifying the reception apparatus **200** of the transmission destination thereof (reception apparatus ID). Then, the flying body **300** stores it in the encryption key storage part **321** as a second encryption key table **150**. The reception apparatus **200** also stores the reception apparatus ID in advance.

[0210] FIG. **15** illustrates one example of the second encryption key table **150**. The second encryption key table **150** is provided with an encryption key recording part **151** and a reception apparatus ID recording part **152**. The encryption key recording part **151** stores the encryption key **500** therein. The reception apparatus ID recording part **152** stores the reception apparatus ID therein. In a case where one encryption key **500** is delivered to a plurality of reception apparatuses **200**, number records of delivery destinations having the same encryption key **500** are registered therein.

[0211] The flying body **300** receives the reception apparatus ID together with an encryption key transmission request from the reception apparatus **200**, at a time of authentication each time it arrives at the reception apparatus **200**. The flying body **300** then extracts all encryption keys **500** stored therein corresponding to the received reception apparatus ID and transmits them to the reception apparatus **200**. Then, the delivered records in the second encryption key table **150** are erased.

[0212] For example, in the example illustrated in FIG. **15**, in a case where the flying body **300** arrives at a reception apparatus **200** whose reception apparatus ID is RCV001, it transmits an encryption key AAA and an encryption key BBB to the reception apparatus **200**. Then, it erases these records. In a case where the flying body **300** arrives at a reception apparatus **200** whose reception apparatus ID is RCV002, it transmits encryption keys AAA and CCC thereto and erases these records. In a case where the flying body **300** arrives at a reception apparatus **200** whose reception apparatus ID is RCV003, it transmits an encryption key CCC thereto and erases this record. In a case where the flying body **300** arrives at a reception apparatus **200** whose reception apparatus ID is RCV0034, it transmits an encryption key AAA thereto and erases this record.

Variation Example 3

[0213] In each of the above example embodiments, a flying body that navigates according to a pre-stored program, such as a drone, is used as an example flying body **300**, but the flying bodies **300** are not limited thereto. For the flying body **300**, any object is available as long as it can move between two points (from the transmission apparatus **100** to the reception apparatus **200**) unmanned. For example, it may be a remotely controllable flying body. It may also be a bullet or artillery shell that does not propel itself, but only moves with initial propulsion, while still being able to control its arrival position. Furthermore, it can be a manned, flyable device such as an airplane, helicopter, or glider. In this case, it is desirable to configure that a pilot is not able to access a storage part where the encryption key **500** is stored.

[0214] The device that delivers the encryption key **500** from the transmission apparatus **100** to the reception apparatus **200** is not limited to flying bodies. It can be an object that can move between two apparatuses which cannot have a transmission line therebetween and that are difficult to access by other objects. For example, it can be a vehicle traveling on the ground, such as an automated vehicle, or a moving body, such as a marine vessel.

Variation Example 4

[0215] It can be configured to earn profit using the encryption key delivery system of each example embodiment or each variation example above.

[0216] In this case, as illustrated in FIG. **16**, an encryption key delivery system **970** is provided with a charging processing apparatus **700** in addition to the configuration of each of the above example embodiments. The case in which the charging processing apparatus **700** is added to the configuration of the first example embodiment is illustrated here. The base encryption key delivery system may be an encryption key delivery system of any other example embodiment or variation example.

[0217] The charging processing apparatus **700** obtains predetermined charging parameters such as the number of navigation courses, delivery distance, the number of deliveries, data volume, non-delivery information, etc., each time the encryption key delivery process by the flying body **300** occurs.

[0218] The number of navigation courses is the number of combinations of the transmission apparatus **100** and the reception apparatus **200**. The number of navigation courses may be received from any of the transmission apparatus **100**, the reception apparatus **200** and the flying body **300**.

[0219] The delivery distance is a distance between the transmission apparatus **100** and the reception apparatus **200** for each delivery. The distance may be calculated from GPS data between the two, for example, or from the flight distance obtained from the flight log of flying body **300**. For example, if the transmission apparatus **100** and/or the reception apparatus **200** moves, it may be the distance at the time of departure or the distance at the time of arrival.

[0220] The number of deliveries is the number of encryption keys **500** delivered. The charging processing apparatus **700** collects, for example, the number of encryption keys **500** transmitted by the transmission apparatus **100** and/or the number of encryption keys **500** received by the reception apparatus **200**.

[0221] The data volume is an amount of data of the encryption key **500** to be delivered. The charging processing apparatus **700** collects, for example, the amount of data transmitted by the transmission apparatus **100** and/or the amount of data received by the reception apparatus **200**.

[0222] The non-delivery information is information that means that the information has been non-delivery. For example, if the flying body **300** erases the encryption key **500** before delivery to the reception apparatus **200**, it also transmits that information to the charging processing apparatus **700**. Alternatively, it may transmit this information via the transmission apparatus **100** or the reception apparatus **200**.

[0223] The charging processing apparatus **700** collects these charging parameters, sums them up for each predetermined period, calculates a charging amount according to predetermined calculation conditions, and charges the predetermined charging destination. The calculation condition may be, for example, a calculation method that multiplies the delivery distance, data volume, and predetermined coefficient of successful deliveries over a predetermined period of time. For example, the calculation method may multiply the delivery distance, data volume, and predetermined coefficient of efficiency for deliveries that occurred regardless of success or failure during a predetermined period, and then subtract a penalty for the number of non-deliveries therefrom. Calculation conditions are not limited to these and can be defined arbitrarily.

[0224] The charging processing apparatus **700** is realized by a general-purpose information processor provided with a CPU, a memory, and a communication interface.

[0225] While each example embodiment of the present disclosure has thus been described, the

present disclosure is not limited thereto. Further modifications, replacements, and adjustments can be made without departing from a basic technical concept of the present invention. For example, configurations of the networks and configurations of each element illustrated in each drawing have been used only as examples to facilitate understanding of the present disclosure. That is, the present disclosure is not limited to the configurations illustrated in the drawings.

[0226] Finally, preferred modes of the present disclosure will be summarized.

[0227] The whole or part of each example embodiment and each variation example disclosed above can be described as, but not limited thereto, the following Modes.

[Mode 1]

[0228] (Refer to the moving body according to the first aspect.)

[Mode 2]

[0229] The moving body above preferably further includes an abnormality detecting part that detects the abnormality and notifies the storage control part.

[Mode 3]

[0230] In the moving body above, it is preferable that the input/output part receives from an external apparatus that the abnormality has been detected and notifies the storage control part.

[Mode 4]

[0231] The moving body above preferably further includes an authentication request part that requests the transmission apparatus and the reception apparatus to authenticate the moving body prior to transmitting and receiving the encryption key.

[Mode 5]

[0232] In the moving body above, it is preferable that the input/output part communicates the encryption key with the transmission apparatus and the reception apparatus using a laser beam.

[Mode 6]

[0233] In the moving body above, it is preferable that the input/output part communicates the encryption key with the transmission apparatus and the reception apparatus using a quantum cryptography communication.

[Mode 7]

[0234] In the moving body above, it is preferable that the abnormality includes at least one selected from the group consisting of an abnormal value, an interference of another object with the moving body, data transmission between the moving body and another device, and an abnormality resulting from analysis of log information of the moving body, and the abnormal value is at least one selected from the group consisting of abnormal values of vibration, temperature, voltage, position, movement speed, and movement time of the moving body.

[Mode 8]

[0235] In the moving body above, it is preferable that the moving body is a flying body.

[Mode 9]

[0236] (Refer to the encryption key delivery system according to the second aspect.)

[Mode 10]

[0237] In the encryption key delivery system above, it is preferable that the transmission apparatus further includes a transmission apparatus authentication part that authenticates the moving body, the reception apparatus further includes a reception apparatus authentication part that authenticates the moving body, the transmission apparatus communication part transmits the encryption key to the moving body in a case where the transmission apparatus authentication part succeeds in authentication, and the reception apparatus communication part receives the encryption key from the moving body in a case where the reception apparatus authentication part succeeds in authentication.

[Mode 11]

[0238] The encryption key delivery system above preferably further includes the moving body according to any one of Modes 1 to 8 as a second moving body, and the second moving body

preferably receives an encryption key from the transmission apparatus, the encryption key received being the same as the encryption key delivered by the moving body.

[Mode 12]

[0239] The encryption key delivery system above preferably further includes the moving body according to any one of Modes 1 to 8 as a second moving body, and the second moving body preferably receives an encryption key from the transmission apparatus, the encryption key received being different from the encryption key delivered by the moving body.

[Mode 13]

[0240] The encryption key delivery system above preferably further includes a charging processing apparatus that collects parameters related to charging generated by the encryption key delivery system and performs charging.

[Mode 14]

[0241] (Refer to the encryption key delivery method according to the third aspect.)

[Mode 15]

[0242] (Refer to the program according to the fourth aspect.) The Modes 14 and 15 can be expanded in the same manner as Mode 1 is expanded to Modes 2 to 8.

[0243] The disclosure of each of above cited PTLs is incorporated herein by reference thereto.

[0244] Modifications and adjustments of the example embodiments or examples are possible within the scope of the overall disclosure (including the claims) of the present disclosure and based on the basic technical concept of the present invention.

[0245] Various combinations and selections of examples and disclosed elements (including the elements in each of the claims, example embodiments, examples, drawings, etc.) are possible within the scope of the claims of the present disclosure.

[0246] That is, the present disclosure includes various variations and modifications that could be made by those skilled in the art according to the overall disclosure including the claims and the technical concept.

[0247] Particularly, any numerical ranges disclosed herein should be interpreted that any intermediate values or subranges falling within the disclosed ranges are also concretely disclosed even without specific recital thereof.

REFERENCE SIGNS LIST

[0248] **100**: transmission apparatus [0249] **111**: (transmission apparatus) communication part [0250] **112**: (transmission apparatus) control part [0251] **113**: (transmission apparatus) authentication part [0252] **121**: encryption key storage part [0253] **122**: authentication information storage part [0254] **131**: CPU [0255] **132**: storage device [0256] **133**: communication device [0257] **140**: encryption key table [0258] **141**: authentication information recording part [0259] **142**: encryption key recording part [0260] **150**: second encryption key table [0261] **151**: encryption key recording part [0262] **152**: reception apparatus ID recording part [0263] **200**: reception apparatus [0264] **211**: (reception apparatus) communication part [0265] **212**: (reception apparatus) control part [0266] **213**: (reception apparatus) authentication part [0267] **221**: encryption key storage part [0268] **222**: authentication information storage part [0269] **231**: CPU [0270] **232**: storage device [0271] **233**: communication device [0272] **300**: flying body [0273] **311**: input/output part [0274] **312**: storage control part [0275] **313**: abnormality detecting part [0276] **314**: authentication request part [0277] **315**: movement control part [0278] **321**: encryption key storage part [0279] **322**: authentication information storage part [0280] **323**: log information storage part [0281] **331**: CPU [0282] **332**: storage device [0283] **333**: communication device [0284] **334**: sensor [0285] **335**: navigation device [0286] **400**: monitor apparatus [0287] **411**: communication part [0288] **412**: information acquisition part [0289] **413**: abnormality detection part [0290] **431**: CPU [0291] **432**: storage device [0292] **433**: communication device [0293] **434**: information collection device [0294] **500**: encryption key [0295] **600**: log information [0296] **700**: charging processing apparatus [0297] **910**: encryption key delivery system [0298] **920**: encryption key delivery system [0299] **940**:

Claims

- 1.** A moving body, comprising: at least a processor; and a memory in circuit communication with the processor, wherein the processor is configured to execute program instructions stored in the memory to implement: an input/output part that receives an encryption key from a transmission apparatus and transmits the encryption key to a reception apparatus; a storage control part that controls storage of the encryption key in a storage part; and a movement control part that moves the moving body to the reception apparatus after receiving the encryption key from the transmission apparatus, wherein the storage control part stores the encryption key in the storage part upon receiving the encryption key from the transmission apparatus and erases the encryption key stored in the storage part in a case where an abnormality is detected.
- 2.** The moving body according to claim 1, wherein the processor is configured to execute program instructions stored in the memory to further implement: an abnormality sensing part that detects the abnormality and notifies the storage control part.
- 3.** The moving body according to claim 1 wherein the input/output part receives from an external apparatus that the abnormality has been detected and notifies the storage control part.
- 4.** The moving body according to claim 1, wherein the processor is configured to execute program instructions stored in the memory to further implement: further comprising: an authentication request part that requests the transmission apparatus and the reception apparatus to authenticate the moving body prior to transmitting and receiving the encryption key.
- 5.** The moving body according to claim 1, wherein the input/output part communicates the encryption key with the transmission apparatus and the reception apparatus using a laser beam.
- 6.** The moving body according to claim 1, wherein the input/output part communicates the encryption key with the transmission apparatus and the reception apparatus using a quantum cryptography communication.
- 7.** The moving body according to claim 1, wherein the abnormality includes at least one selected from the group consisting of an abnormal value, an interference of another object with the moving body, data transmission between the moving body and another device, and an abnormality resulting from analysis of log information of the moving body, wherein the abnormal value is at least one selected from the group consisting of abnormal values of vibration, temperature, voltage, position, movement speed, and movement time of the moving body.
- 8.** The moving body according to claim 1, wherein the moving body is a flying body.
- 9.** An encryption key delivery system, comprising: a first moving body that is the moving body according to claim 1; a transmission apparatus; and a reception apparatus, wherein the transmission apparatus comprises at least a processor; and a memory in circuit communication with the processor, wherein the processor is configured to execute program instructions stored in the memory to implement: a transmission apparatus communication part that transmits the encryption key to the moving body, and the reception apparatus comprises at least a processor; and a memory in circuit communication with the processor, wherein the processor is configured to execute program instructions stored in the memory to implement: a reception apparatus communication part that receives the encryption key from the moving body.
- 10.** The encryption key delivery system according to claim 9, wherein the transmission apparatus further comprises a transmission apparatus authentication part that authenticates the moving body, the reception apparatus further comprises a reception apparatus authentication part that authenticates the moving body, the transmission apparatus communication part transmits the encryption key to the moving body in a case where the transmission apparatus authentication part succeeds in authentication, and the reception apparatus communication part receives the encryption

key from the moving body in a case where the reception apparatus authentication part succeeds in authentication.

11. The encryption key delivery system according to claim 9, further comprising: a second moving body having a same configuration as the first moving body; wherein the second moving body receives an encryption key from the transmission apparatus, the encryption key received being the same as the encryption key delivered by the first moving body.

12. The encryption key delivery system according to claim 9, further comprising: a second moving body having a same configuration as the first moving body; wherein the second moving body receives an encryption key from the transmission apparatus, the encryption key received being different from the encryption key delivered by the first moving body.

13. The encryption key delivery system according to claim 9, further comprising: a charging processing apparatus that comprises at least a processor; and a memory in circuit communication with the processor, wherein the processor is configured to: collect parameters related to charging generated by the encryption key delivery system and performs charging.

14. An encryption key delivery method executed by a computer loaded on a moving body, the encryption key delivery method, comprising: receiving an encryption key from a transmission apparatus; storing the encryption key received in a storage part; moving the moving body toward a reception apparatus after storing the encryption key; erasing the encryption key stored in the storage part in a case where an abnormality of the moving body is detected during a movement; and transmitting the encryption key stored in the storage part to the reception apparatus, upon arriving at a destination area.

15. A non-transitory computer-readable medium storing a program causing a computer loaded on a moving body to execute; receiving an encryption key from a transmission apparatus; storing the encryption key received in a storage part; moving the moving body toward a reception apparatus after storing the encryption key; erasing the encryption key stored in the storage part in a case where an abnormality of the moving body is detected during a movement; and transmitting the encryption key stored in the storage part to the reception apparatus, upon arriving at a destination area.

16. The encryption key delivery method according to claim 14, further comprising: receiving from an external apparatus that the abnormality has been detected and notifying.

17. The encryption key delivery method according to claim 14, further comprising: requesting the transmission apparatus to authenticate the moving body prior to transmitting the encryption key, and requesting the reception apparatus to authenticate the moving body prior to receiving the encryption key, upon arriving at the destination area.

18. The encryption key delivery method according to claim 14, wherein a reception of the encryption key from the transmission apparatus and a transmission of the encryption key to the transmission apparatus are performed using a laser beam.

19. The encryption key delivery method according to claim 14, wherein a reception of the encryption key from the transmission apparatus and a transmission of the encryption key to the transmission apparatus are performed using a quantum cryptography communication.

20. The encryption key delivery method according to claim 14, wherein the abnormality includes at least one selected from the group consisting of an abnormal value, an interference of another object with the moving body, data transmission between the moving body and another device, and an abnormality resulting from analysis of log information of the moving body, wherein the abnormal value is at least one selected from the group consisting of abnormal values of vibration, temperature, voltage, position, movement speed, and movement time of the moving body.
